3. Core + · Write up [DEM] The equivalence " $x^2 = 4 \Leftrightarrow x = 2$" correct? Correct it.

Ans. No: $x = -2$ also works. We have the implication $x = 2 \Rightarrow x^2 = 4$ and the equivalence $x^2 = 4 \Leftrightarrow (x = 2 \text{ or } x = -2)$.

4. Back · Flash [CMP] Compare 2^{10} and 10^3.

Ans. $2^{10} = 1\,024 > 1\,000 = 10^3$.

5. Running thread · Flash [REP] Between which consecutive integers does $\frac{7}{4}$ lie on a number line? Give its decimal value.

Ans. Between 1 and 2; 1.75.

Sheet 61 Mon. 26 Apr. 2027 Oral class discussion: comparing methods

1. Warm-up · Flash [GEO] Distance between $A(-2;1)$ and $B(1;5)$?

Ans. $\sqrt{3^2 + 4^2} = 5$.

2. Core · Method [GEO] Give three methods to prove that A, B, C are collinear.

Ans. (i) $\det\left(\overrightarrow{AB}, \overrightarrow{AC}\right) = 0$; (ii) equation of (AB) satisfied by C ; (iii) $AB + BC = AC$: equal distances, equivalent to $B \in [AC]$.

3. Core + · Reasoning [GEO] To prove that two lines given by $y = m_1 x + p_1$ and $y = m_2 x + p_2$ are parallel, which method is quicker?

Ans. Compare the slopes: $m_1 = m_2$; strictly parallel if, moreover, $p_1 \neq p_2$.

4. Back · Factorise [DEV] Factorise $6x + 9$.

Ans. $3(2x + 3)$.

5. Running thread · Method [DIV] Write 360 as a product of primes. How many divisors does it have?

Ans. $360 = 2^3 \times 3^2 \times 5$; $(3+1)(2+1)(1+1) = 24$ divisors.

Synthesis problems, reasoning and fluency

Sheet 62 Tue. 27 Apr. 2027 Synthesis problems spanning several areas of the curriculum

1. Warm-up · Flash [EQ] Solve $x - 3 = 2x + 9$.

Ans. $3x = 12$, $x = 4$.

2. Core · Problem [PRO] A price undergoes a 20 % rise then a fixed discount of 1,000 Ar and is then 11,000 Ar. What was the original price?

Ans. $1{,}2p - 1\,000 = 11\,000$, $p = \frac{12\,000}{1{,}2} = 10{,}000$ Ar.

3. Core + · Method [REG] $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the intersection point of the curves then solve $f(x) < g(x)$.

Ans. $2x + 1 = -x + 7$: $x = 2$, point $(2;5)$; $f(x) < g(x) \Leftrightarrow 3x < 6 \Leftrightarrow x < 2$.

4. Back · Flash [QP] A recipe for 4 people requires 250 g of flour. How much for 6 people?

Ans. $250 \div 4 \times 6 = 375$ g.

5. Running thread · Flash [GEO] A figure with area 4 cm^2 is enlarged by a scale factor of 3. What is the area of the enlargement?

Ans. Area $\times 3^2$: 36 cm^2.

Pitfall: multiplying the area by 3 instead of 3^2.